KENYA INSTITUTE OF CURRICULUM DEVELOPMENT
A Skilled and Ethical Society

SENIOR SCHOOL CURRICULUM DESIGN

ESSENTIAL MATHEMATICS
GRADE 11

OCTOBER, 2025

DRAFT

First Published in 2025

All rights reserved. No part of this book may be reproduced, stored in a retrieval system or transcribed, in any form or by any means, electronic, mechanical, photocopy, recording or otherwise, without the prior written permission of the publisher.

ISBN: 978-9914-43-068-4

Published and printed by Kenya Institute of Curriculum Development

TABLE OF CONTENTS

TABLE OF CONTENTS i
NATIONAL GOALS OF EDUCATION ii
THE SENIOR SCHOOL IN THE COMPETENCY BASED CURRICULUM iv
LEARNING OUTCOMES FOR SENIOR SCHOOL iv
TABLE 1: LIST OF SUBJECTS AT SENIOR SCHOOL v
TABLE 2: LESSON DISTRIBUTION AT SENIOR SCHOOL vi
ESSENCE STATEMENT vii
SUBJECT GENERAL LEARNING OUTCOMES viii
STRANDS viii
SUMMARY OF STRANDS AND SUB STRANDS ix
STRAND: NUMBERS AND ALGEBRA 1
STRAND 2.0: MEASUREMENTSS AND GEOMETRY 10
STRAND 3.0: STATISTICS AND PROBABILITY 23
APPENDIX 1: SUMMARY OF RESOURCES, ASSESSMENT METHODS AND NON-FORMAL ACTIVITIES 28
APPENDIX 2: USE OF ICT TOOLS 31

NATIONAL GOALS OF EDUCATION

Education in Kenya should:

1. **Foster nationalism and patriotism and promote national unity.**
   Kenya's people belong to different communities, races and religions, but these differences need not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationhood by resolving conflicts and promoting positive attitudes of mutual respect, which enable them to live together in harmony and foster patriotism in order to make a positive contribution to the life of the nation.

2. **Promote the social, economic, technological and industrial needs for national development.**
   Education should prepare the youth of the country to play an effective and productive role in the life of the nation.
   a) **Social Needs**
      Education in Kenya must prepare children for changes in attitudes and relationships which are necessary for the smooth progress of a rapidly developing modern economy. There is bound to be a silent social revolution in the wake of rapid modernisation. Education should assist our youth to adapt to this change.

NATIONAL GOALS OF EDUCATION

Education in Kenya should:

1. Foster nationalism and patriotism and promote national unity.
   Kenya's people belong to different communities, races and religions, but these differences need not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationhood by resolving conflicts and promoting positive attitudes of mutual respect, which enable them to live together in harmony and foster patriotism in order to make a positive contribution to the life of the nation.

2. Promote the social, economic, technological and industrial needs for national development.

NATIONAL GOALS OF EDUCATION

Education in Kenya should:

1. Foster nationalism and patriotism and promote national unity.
   Kenya's people belong to different communities, races and religions, but these differences need not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationhood by resolving conflicts and promoting positive attitudes of mutual respect, which enable them to live together in harmony and foster patriotism in order to make a positive contribution to the life of the nation.

2. Promote the social, economic, technological and industrial needs for national development.
   Education should prepare the youth of the country to play an effective and productive role in the life of the nation.
   a) Social Needs
      Education in Kenya must prepare children for changes in attitudes and relationships which are necessary for the smooth progress of a rapidly developing modern economy. There is bound to be a silent social revolution in the wake of rapid modernisation. Education should assist our youth to adapt to this change.

   b) **Economic Needs**
      Education in Kenya should produce citizens with the skills, knowledge, expertise and personal qualities that are required to support a growing economy. Kenya is building a modern and independent economy which is in need of an adequate and relevant